

CLASS 12 / NOV 10 / U09

DISCRETIZING GEOMETRY

P4 CONTINUES — FROM A CONTINUOUS VOLUME TO ORDERED PARTS

P4a DUE

P4b ISSUED TODAY

TODAY

Three things change hands before we cut a single section.

DUE

P4a — SDF VOLUME

DUE WEEK 12 — Nov 10 / 14%

Two sheets as PDF, plus the editable field graph with the source fields retained.

CHECKPOINT

BOOK — PROGRESS CHECK

WEEK 12 — Nov 10

Bring a booklet spread and contents list for an ungraded progress check. Ungraded.

ISSUED

P4b — SLICES

DUE FINAL / 14%

Your P4a volume becomes laser-cut parts. Brief at the end of the class.

THE VOLUME YOU SELECTED ON P4a SHEET 2 IS THE ONE YOU SLICE. DO NOT START A NEW ONE.

FOUR DECISIONS BEFORE ANY CUT

A slice set is not a setting. Each of these changes what the assembly says about the volume.

PLANE

What cuts the volume: a plane, repeated. The contour is where they meet.

DIRECTION

Which way the planes face. It decides which profiles you see.

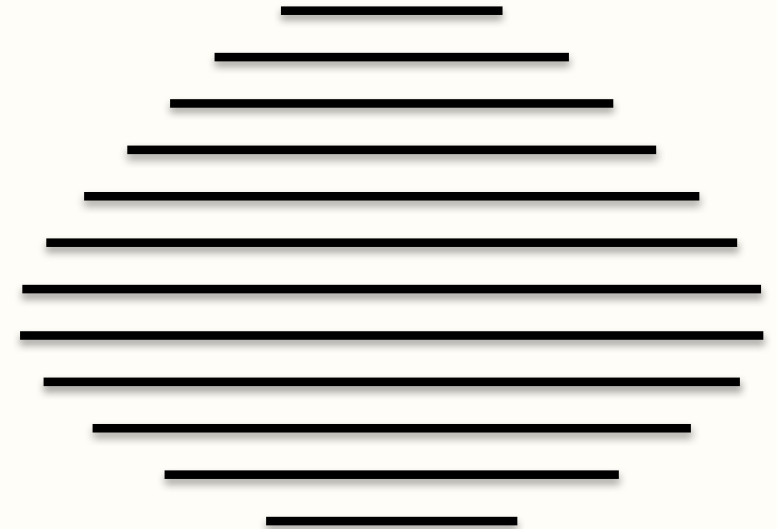
INTERVAL

The distance between planes. It sets the gaps, the light and the part count.

ORDER

Which slice is first. The order becomes the labels and the assembly.

ONE VOLUME, 12 HORIZONTAL PLANES



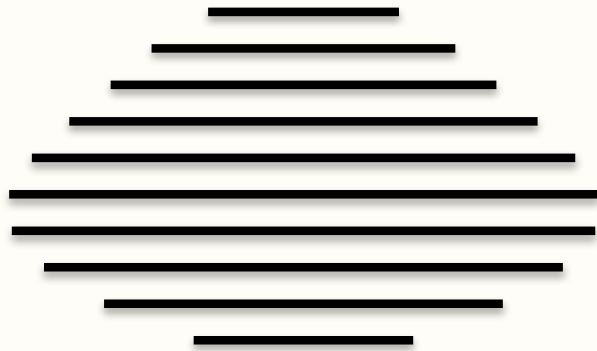
01 at the bottom · 12 at the top

P4b asks for at least 12 ordered slices. Twelve is the floor, not the target.

COMPARE BEFORE YOU COMMIT

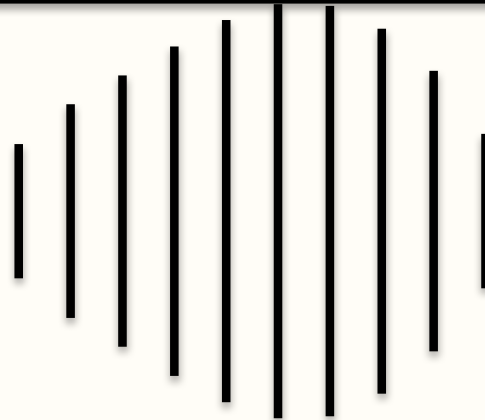
Same volume, three slice sets. P4b asks you to compare two directions or two intervals.

HORIZONTAL · WIDE



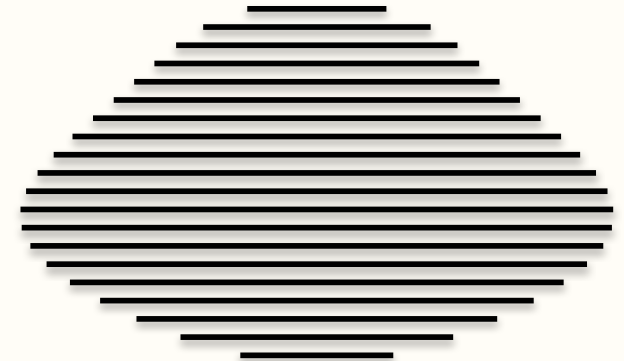
Reads as a stack. Clear profile, open gaps.

VERTICAL · WIDE



Reads as fins. The silhouette changes with view.

HORIZONTAL · TIGHT



Nearly continuous. More parts, less light.

LOST

Everything between the planes. Curvature becomes steps.

GAINED

Gaps, light and a reading of the form. Parts a machine can cut.

SLICES ARE DESIGNED PARTS

In Week 3 a slicer made the layers for you. Here, every section is a part you decide on.

PRINTER LAYERS

Generated by the slicer, not by you.

Hundreds of them, fused into one object.

You never see one on its own.

SECTION PARTS

Chosen: direction, interval, order.

A countable set, each with a thickness, a label and a place in the assembly.

Every one is visible in the finished piece.

DECIDE WHICH CURVES ARE PART BOUNDARIES BEFORE YOU ASSIGN THICKNESS OR SEND A MACHINE FILE.

THE SEQUENCE

Steps 1 and 2 today. Steps 3 and 4 are next week's class.

01

Intersect the volume or its extracted boundary with a sequence of planes; order the resulting contour curves.

02

Compare two section directions and intervals, including the relationship between gaps, light and the reading of the form.

03

Close and clean the intended profiles; turn them into parts with material thickness and a clear registration system.

04

Test fit and kerf, then number, nest and fabricate the parts and document the assembly.

THE SAME THING, WRITTEN OUT

Slice in order, then prepare each part. The loop is the same one you wrote for arrays.

```
sections = []
for k in range(count):
    z = z0 + k * interval
    sections.append(slice(volume, z))    # ordered, and order matters

for i, s in enumerate(sections):
    s = offset(s, kerf / 2)             # the blade has width
    label(s, i)
    nest(s, sheet)
```

WHAT IT MAPS ONTO

section plane
contour
interval
discretization
thickness
registration
kerf
nesting

THE FIRST LOOP IS TODAY. THE SECOND LOOP — KERF, LABEL, NEST — IS NEXT WEEK.

ISSUED TODAY

P4b — DISCRETIZING AND LASER-CUT FABRICATION

DUE FINAL / 14%

Take the volume you selected in P4a and translate it into at least twelve ordered sections. Compare two directions or intervals, give the slices a thickness and a registration system, then laser-cut and assemble them within one sheet of material.

COMPARE

TWO SLICE SETS

Two directions or two intervals, from the same volume.

MAKE

12+ PARTS

Thickness, registration, labels, nesting. Within 300 mm.

SHOW

ONE PROCESS SHEET

Volume, sections, cutting layout, assembly, photographs.

REQUIREMENTS — P4b

Visual documentation 20% · computational understanding 20% · technical execution 15% · fabrication quality 30% · experimentation 5% · requirements and identity 10%.

01

Use the selected P4a volume. Compare two section directions or intervals, then develop at least 12 ordered slices with material thickness and a registration system.

02

Fit the assembly within 300 mm on each axis and one confirmed material sheet. Test kerf and fit before the batch; label and nest every part.

03

Laser-cut and assemble the slices under FabLab procedures. Include test cuts and failed parts in the material allocation.

04

Present the assembly with one 17 x 11 inch process sheet showing the continuous volume, section sequence, cutting layout, assembly diagram and photographs.

05

Submit the editable source, cutting file and documentation. Regenerate fabrication files after any geometry or material change.

CHECKPOINT · WEEK 13 — Nov 17 — Review section directions, thickness, registration, fit/kerf test and cutting layout before production.

NOW — CUT THE VOLUME

Rest of the class: sections from your P4a volume, compared before anything is chosen.

✓

P4a submitted, and your booklet spread and contents list checked

✓

Your selected volume sectioned by a sequence of planes

✓

The contours ordered, with slice 01 identified

✓

A second slice set: another direction or another interval

✓

The two sets set side by side, with a sentence on what each one shows

IF THE TWO SETS LOOK THE SAME, CHANGE MORE.

BEFORE NEXT CLASS

NEXT

Week 13 (target Nov 17) — Parts for the laser cutter: registration, kerf and fit, labels and nesting.

BRING

A candidate slice set: at least 12 ordered contours, with the direction and interval written down, and your comparison of two sets.

CHECK

P4b checkpoint, Week 13 — Nov 17 (target): Review section directions, thickness, registration, fit/kerf test and cutting layout before production.

FIND

Your material: measure the sheet you plan to use and confirm its size and thickness with the FabLab before you design parts around it.